

How we Tested

The test bed was a Pentium 4, 3.2 GHz system with an Intel 875 motherboard, 512 MB 333 MHz DDR RAM, a GeForce4 Ti4600 graphics card and a 40 GB Seagate 7200 rpm Serial ATA hard drive, running Windows XP Professional.

We rated all the MP3 players according to their features, performance, ease of use, ergonomics, and their value for money.

Features

Features are what make a device more desirable. They make usability and operability of the device easier. We took note of all the features supported by the MP3 players such as battery life, memory size, its interface, backlight, etc. Some important features are explained below:

Memory size: The higher the memory capacity, the more songs you can store on it.

Battery type and number of batteries: MP3 players, especially those with backlit LCD screens, are power hogs. Those with internal rechargeable batteries are preferred. Also, those that require only a single battery to operate are preferred over those that require two or more.

LCD screen and backlight: An LCD is an obvious advantage as it displays the track, artist and audio format being played. Browsing the menu is also a lot easier.

Remote control: A remote control brings obvious benefits and conveniences.

PC interface: Transferring music onto the player is an important aspect of owning a portable MP3 player. Parallel and serial interfaces are outdated, so those with the faster USB 1.1, USB 2.0 and FireWire interfaces got more points.

Audio formats supported: MP3 isn't the only compressed music format anymore. We awarded points according to the number of formats supported—MP3, WMA, OGG, etc.

Firmware upgrade: Firmware is the software that sits on the chip, and is comparable to an OS. If a player's firmware can be upgraded, it can be relatively future-proofed by adding more features with newer versions of its firmware.

Voice recording: Voice recording capability can make your player double as a dicta-phone. This is obviously desirable.

AM and FM Tuner: An in-built radio tuner offers more variety, and you will not have to make do with only what's on your player—especially when you haven't had the time to load new songs.

ID3 tag: An ID3 tag is basically 128 bytes of track information stored at the beginning of a song—track name, artist, genre, etc. Devices that can read and display this information were awarded more points.

Track Browsing: Track browsing is a much needed feature which lets you jump to any song, and does not limit you to countless clicking of the forward or rewind buttons.

Data Storage: Certain MP3 players can double as portable storage devices. This is an added advantage.

Line-Out: A Line-Out connection allows you to plug your player to

other devices, such as your home or car audio system.

Bundled software and accessories: Free software is always welcome! We noted the software bundled and its usefulness, and also made a note of the accessories provided, such as a carry pouch, arm band etc.

Equaliser: An equaliser lets you tweak the playback to suite your tastes. We noted the number of presets available, and whether you can customise the bass and treble levels.

Playlist feature: A playlist lets you choose which tracks you want to listen to, and in the order you want to hear them—obviously a boon.

Repeat: The repeat feature lets you hear your favourite song again and again and again and again and again...

Shuffle: This lets you play songs in random order, for those of you who love surprises.

Performance

What's the use of a player with tons of features, but performs badly. Here's a list of things we kept an eye out for.

Audio quality: We tested the MP3 players by playing five tracks—one with high bass, one with high treble, one English track and two Hindi tracks. Though we used a reference headphone set to test all the players, we also used the earphones supplied with each player to test the quality of both the player and its earphones.

Data transfer speed: We noted the time taken to transfer 50 MB set of MP3 files from the test PC to the player using the bundled software, where applicable. The faster the transfer, the more points a player earned.

Playtime or battery-life: The playtime was tested by fully recharging batteries, or inserting new batteries (as applicable), and playing music continuously at full volume until the battery was drained. The players were then awarded points in order of battery performance.

Ease of use and ergonomics

It is very important for portable devices to be user-friendly. The buttons should be easily reachable and simple to press. Preference was given to players with simple yet functional buttons, so that once you're used to the player you will not need to remove it from your pocket to control it. An easy-to-use menu structure is a must, where you will not need more than three clicks to access any function. We also looked for players with a backlit LCD, for easier operation in the dark.

Value for money

This is calculated by dividing the feature and performance offered with the price that the player is available for. This is especially important, as it gives you a clearer picture about whether a player is priced correctly as per the features and performance it offers—the all-important buying decision depends greatly on this.

button placement. However, as there is no hold-button, if you carry the player in your pocket you are likely to press some buttons inadvertently.

The bundled earphones were extremely clear. However, its treble was sharp and using the player for long periods is a

painful experience. The time taken to transfer 50 MB of music to the device over the USB 1.1 interface was a slow 3 minutes and 38 seconds.

Overall, the portability, user-friendliness and good audio output are the definite plus points of this player. At Rs 5,999, the